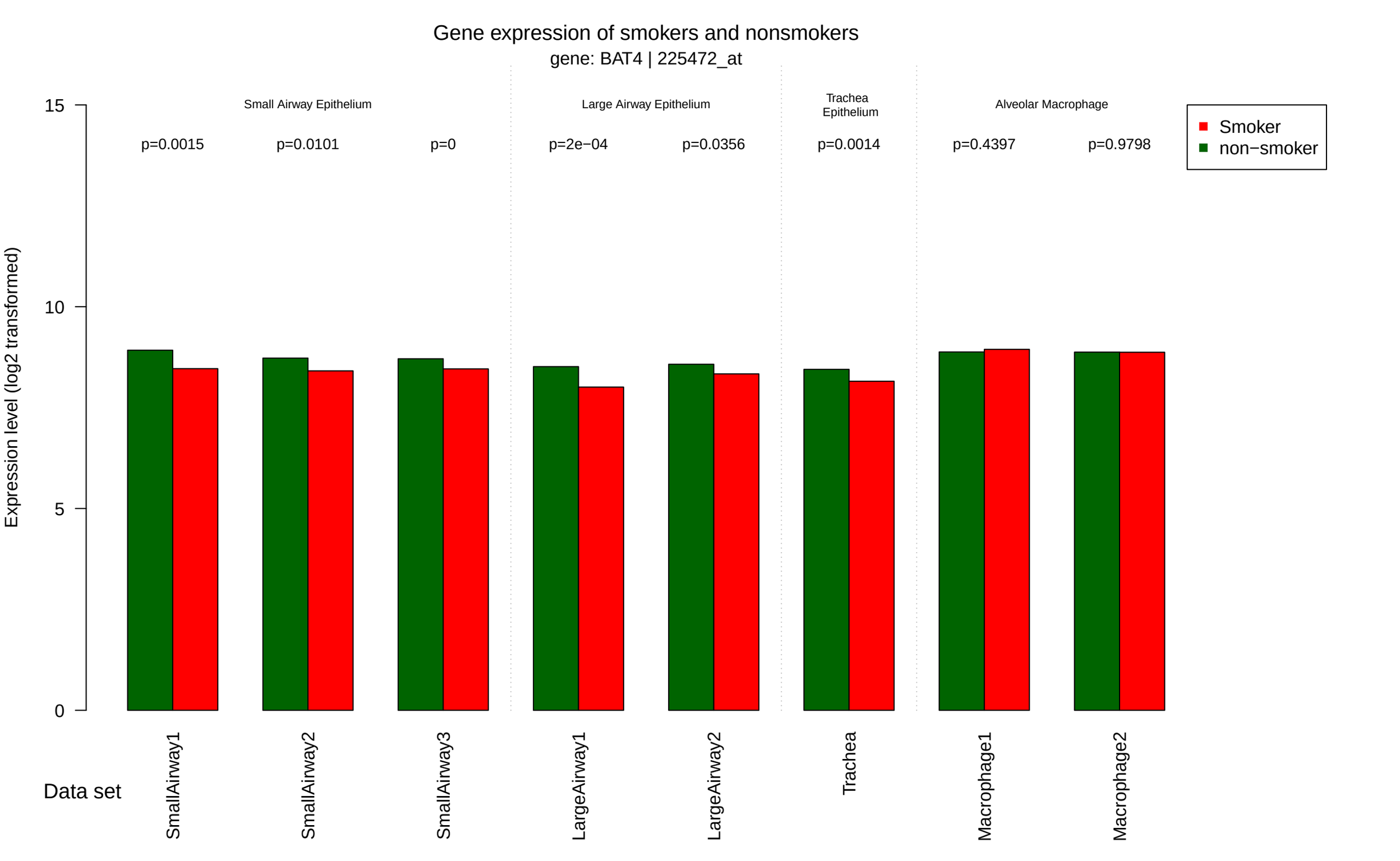
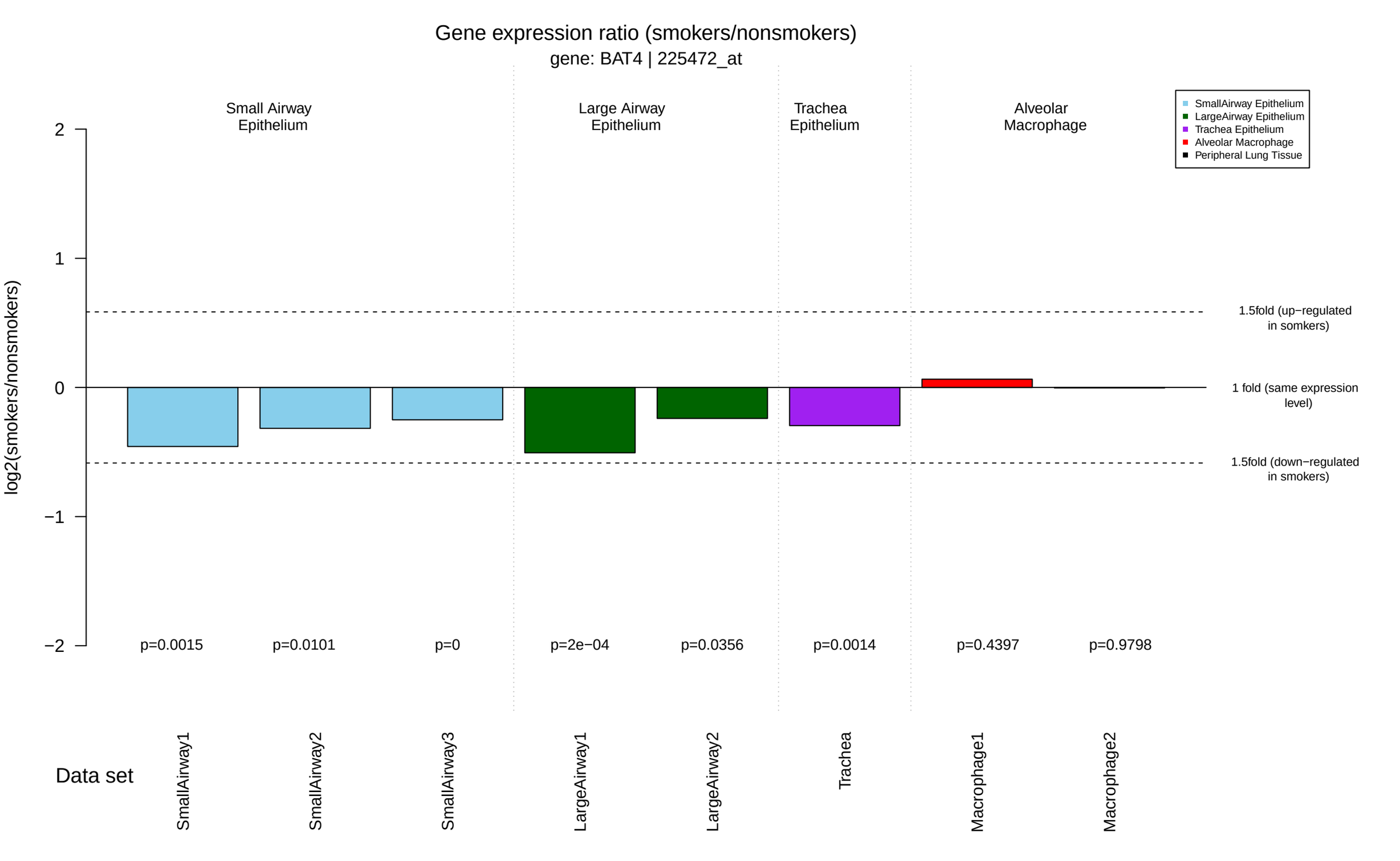
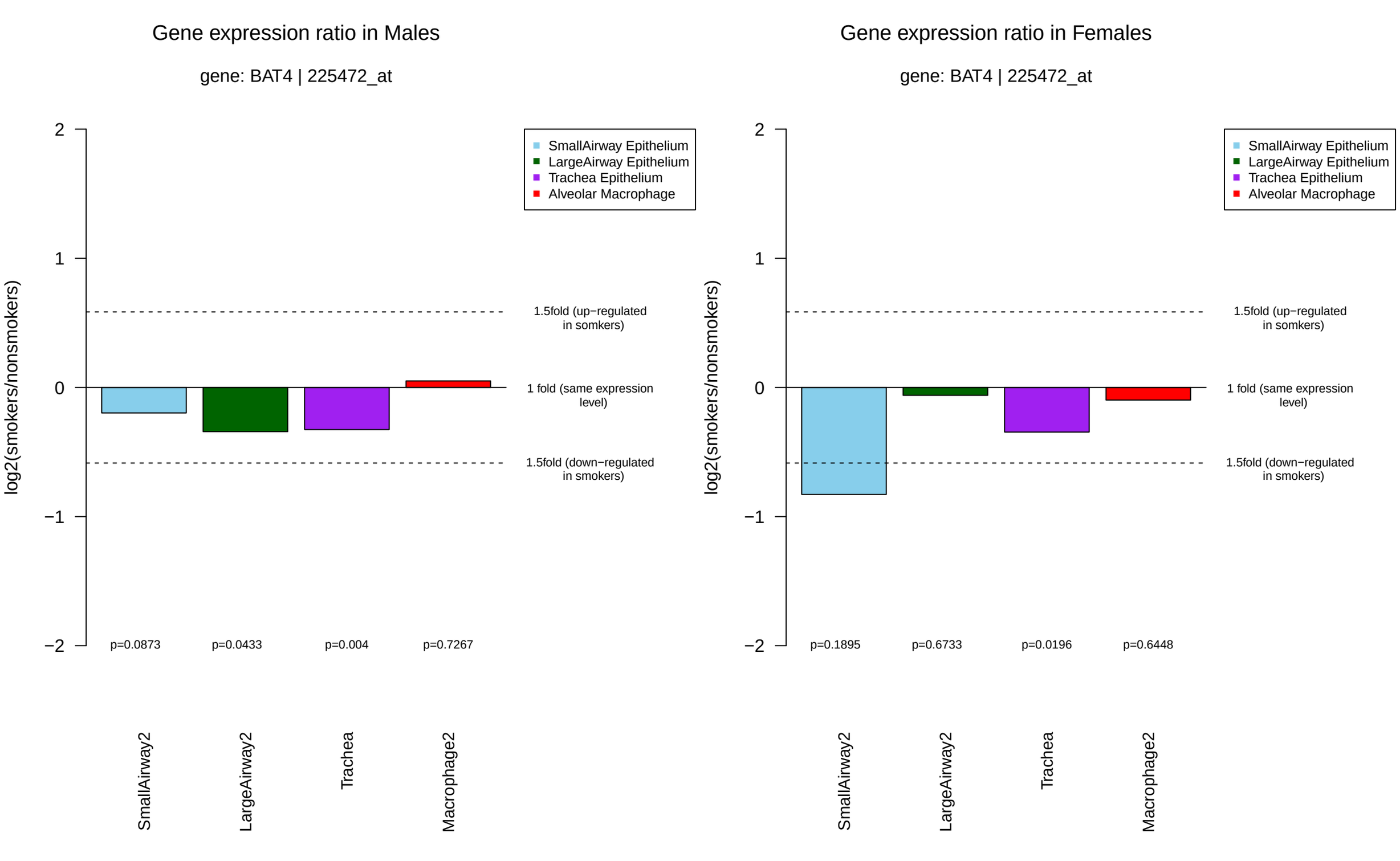
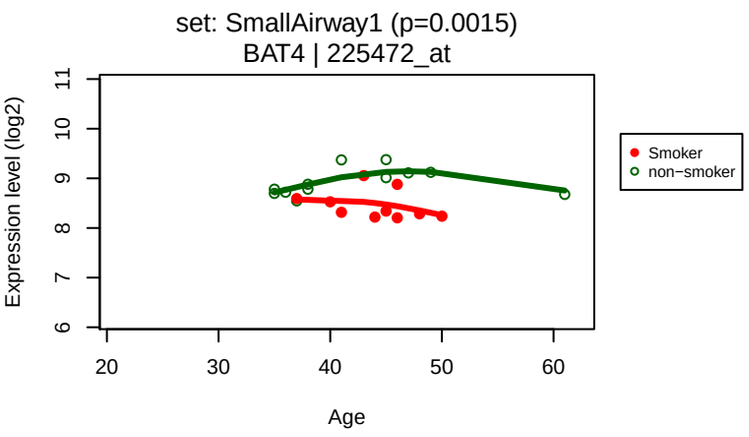
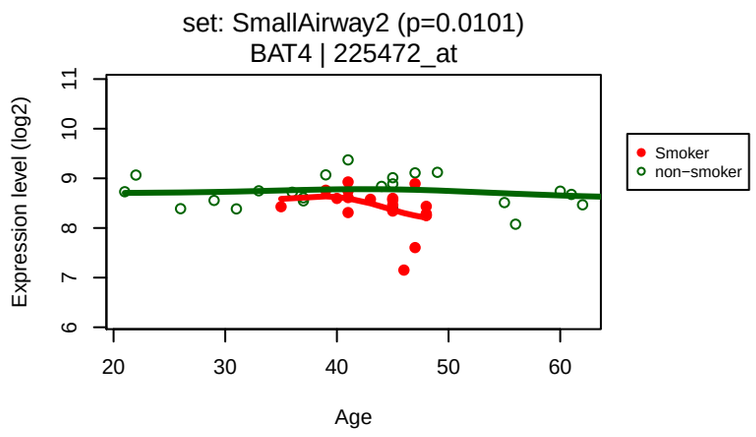
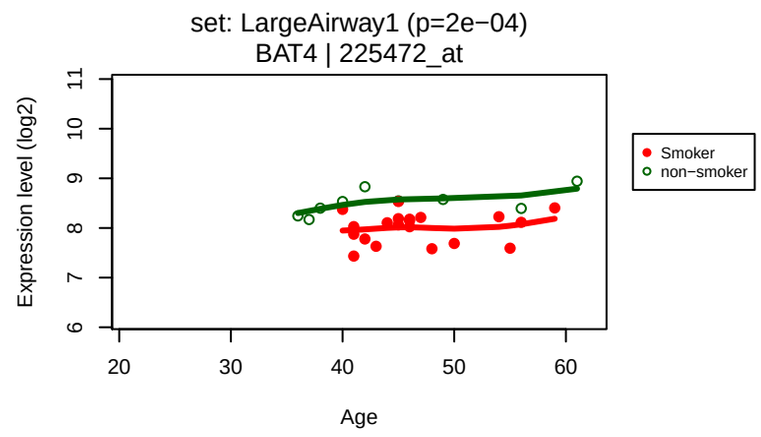

Gene expression of smokers and nonsmokers
gene: BAT4 | 225472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0015
p=0.0101
p=0
p=2e−04
p=0.0356
p=0.0014
p=0.4397
p=0.9798
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

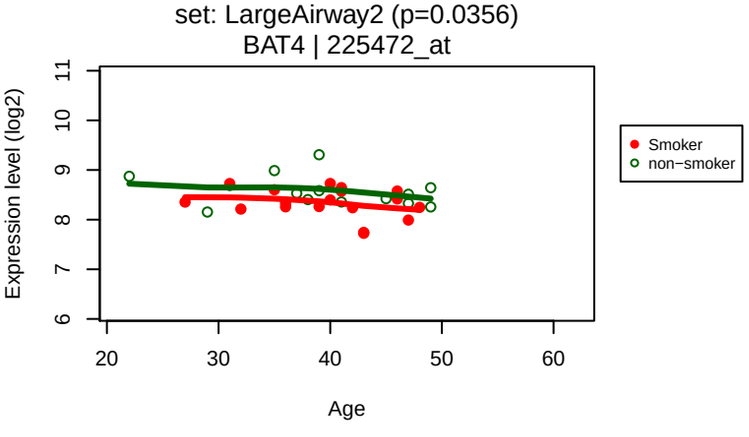

Gene expression ratio (smokers/nonsmokers)
gene: BAT4 | 225472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0015
p=0.0101
p=0
p=2e−04
p=0.0356
p=0.0014
p=0.4397
p=0.9798
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

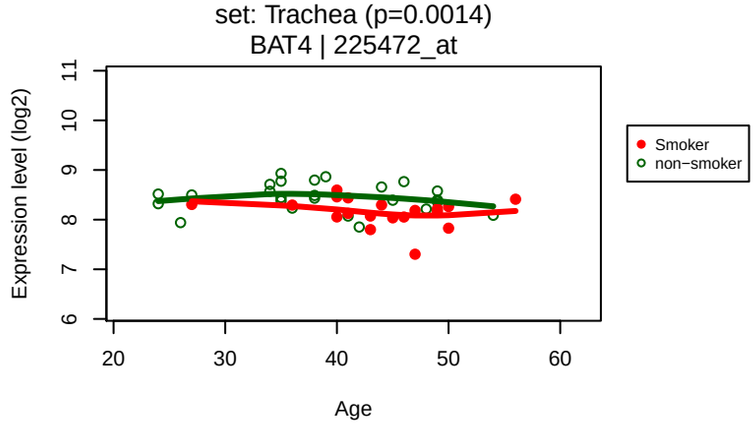

Gene expression ratio in Males
gene: BAT4 | 225472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0873
p=0.0433
p=0.004
p=0.7267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAT4 | 225472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1895
p=0.6733
p=0.0196
p=0.6448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

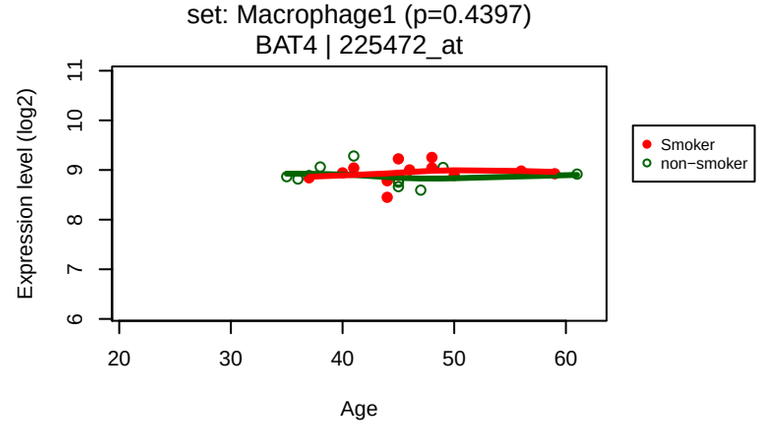

set: SmallAirway1 (p=0.0015)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker

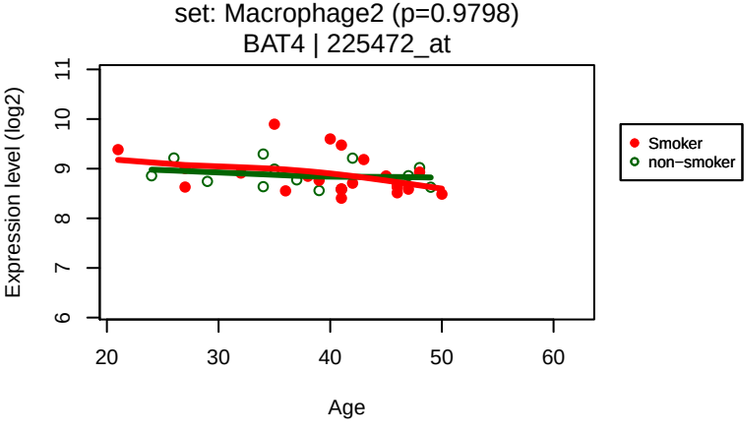

set: SmallAirway2 (p=0.0101)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker

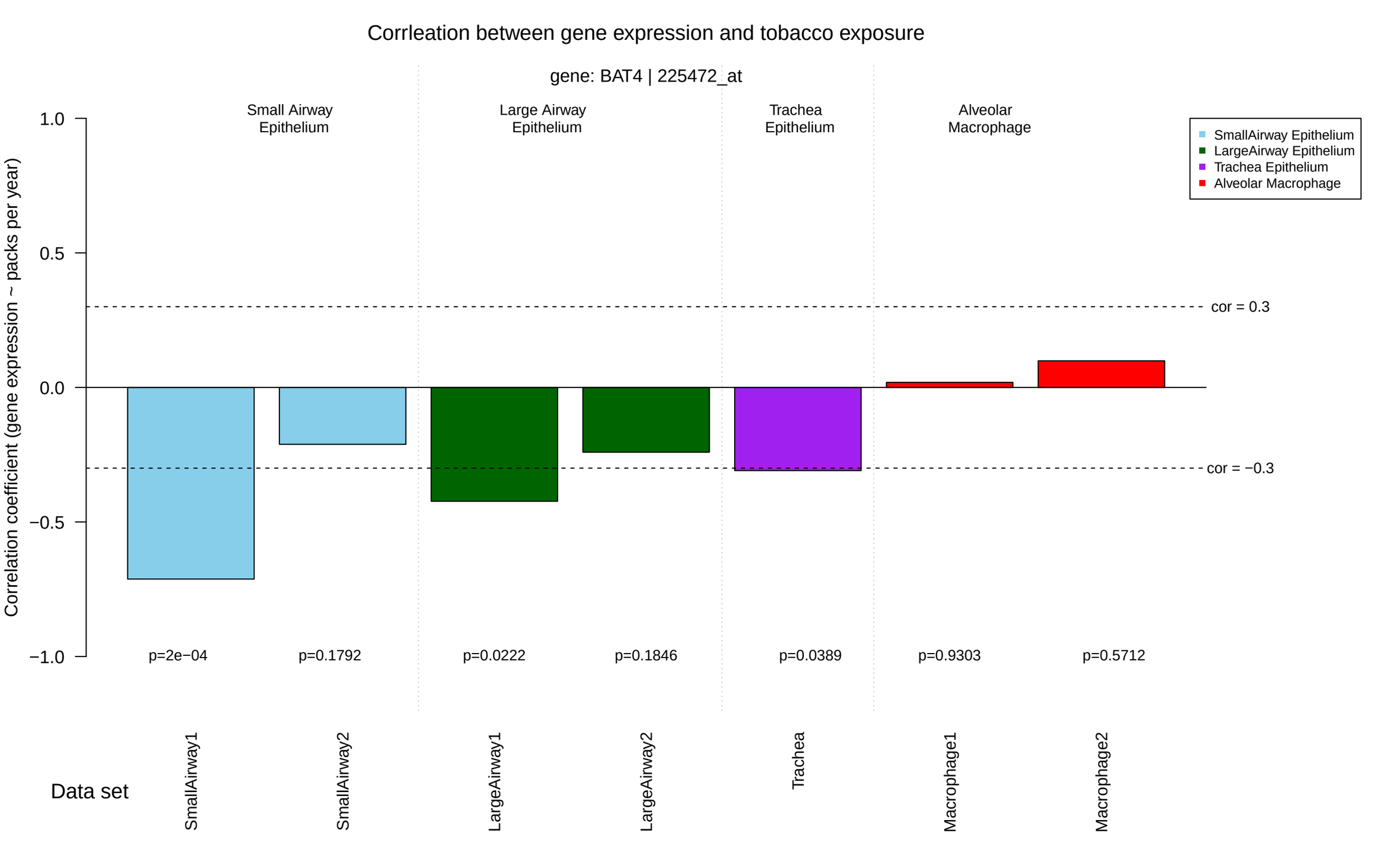

set: LargeAirway1 (p=2e−04)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0356)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4397)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9798)
BAT4 | 225472_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAT4 | 225472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=2e−04
p=0.1792
p=0.0222
p=0.1846
p=0.0389
p=0.9303
p=0.5712
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2